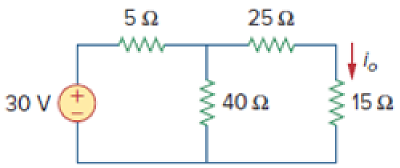
Chapter 4, Problem 1P

Problem

Calculate the current *io* in the circuit of Fig. 4.69. What value of input voltage is necessary to make *io* equal to 5 amps?

Figure. 4.69:



Step-by-step solution

Step 1 of 3

Refer to Figure 4.69 in the textbook.

Redraw the circuit diagram showing with loop currents are shown in Figure 1.

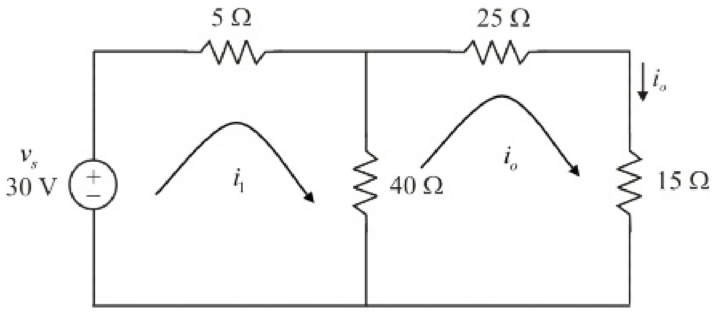


Figure 1

Step 2 of 3

Apply Kirchhoff's voltage law to loop 1 in Figure 1.

$$-v_s+5i_1+40(i_1-i_o)=0$$

$$45i_1-40i_o=v_s \quad \ldots\ldots (1)$$

Apply Kirchhoff's voltage law to loop 2 in Figure 1.

$$40(i_o-i_1)+25i_o+15i_o=0$$
$$80i_o-40i_1=0$$
$$i_1=2i_o$$

Substitute $2i_o$ for i_1 in equation (1).

$$45(2i_o)-40i_o=v_s$$

$$50i_o=v_s \quad \ldots\ldots (2)$$

Substitute 30 V for v_s in equation (2).

$$50i_o=30$$
$$i_o=\frac{30}{50}$$
$$=600\text{ mA}$$

Therefore, the current i_o, is $\boxed{600\text{ mA}}$.

If $i_o = 5\,\text{A}$, the input voltage is,

From Equation (2)

$$\begin{aligned} v_s &= 50(5) \\ &= 250\,\text{V} \end{aligned}$$

Therefore, the required input voltage v_s, is $\boxed{250\,\text{V}}$.